crackling

MATLAB style guide

Cases

variables	`snake_case` (lower case with underscores)
functions	`mixedCase` (initially lower, then upper, without underscores)
objects	`UpperCase` (upper throughout, without underscores)
object properties	`snake_case` (lower case with underscores)
object methods	`mixedCase` (initially lower, then upper, without underscores)

Comment headers

Objects

```
% ObjectName
%
% Simple description of object in full sentences.
% Can be split over multiple lines.
%
% Syntax:
% Output = namespace.ObjectName(Input1, Input2, …)
%
% Any extra details required are included after syntax.
% Again, must be in sentences over multiple lines.
```

Functions and object methods

Function and method header comments are placed between the function declaration line (function …) and the first executable line of code (and before the arguments block).

```
% functionName
%
% Simple description of object in full sentences.
% Can be split over multiple lines.
%
% Syntax:
% [Output1, Output2,…] = namespace.object.functionName(Input1, Input2)
%
% Inputs:
```

```
% Input1 - (size of Input1) class of Input1, description of Input1.
% Input2 - (size of Input2) class of Input2, description of Input2.

%
% Outputs:
% Output1 - (size of Output1) class of Output1, description of Output1
% Output1 - (size of Output1) class of Output1, description of Output1
```

Object properties

```
properties
        % Description of property above declaration
        property (size) class {validators}
end
```

Formatting

- Surround operators with spaces

```
X = Y, X > Y, A && B, etc
```

- Follow commas with spaces

```
foo(a, b, c)
```

- Describe new variables when they are first created

```
% comments can either be above a new variable
new_variable % or beside a new variable
```

- Separate blocks of code with an editor section and two blank lines
- Then separate logical parts of code inside a section with one blank line

```
%% The current section
foo(a, b, c)
bar(x, y, z)

% Single comments can accompany short logical parts
foo2(a, b, c)
bar2(x, y, z)
```

- Do not exceed 80 characters per line

```
% Comments and code can be wrapped around at the end of a line while
continuing sentences and ideas.
```

Property and method attributes

Object property and method attributes are useful ways to describe the way that these should (or should not) be used. They also make clear the important parts of the interface to an object. Where relevant, the following should be used:

Property attributes

Attribute	**Meaning**
Abstract	The property has no implementation, but a concrete subclass must override this property without Abstract set to true.
Constant	This property has only one value in all instances of the class.
Dependent	The property value is not stored in the object. The `set` and `get` functions cannot access the property by indexing into the object using the property name.

Method attributes

Attribute	**Meaning**
Abstract	The method has no implementation. The method has a syntax line that can include arguments that subclasses use when implementing the method.
Static	A method that does not depend on an object of the class and does not require an object argument.